



Azure Estate Report

Quarterly estate review

Contoso Ltd

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Executive assessment

The collected estate contains 52 resources in 2 subscriptions. Its 35 finding occurrences come from 21 recurring checks and refer to 24 distinct resource IDs. Review the recurring configuration patterns and their concentrations before assigning work to individual resources.

52

Resources

21

Recurring checks

24

Affected resources

Principal observations

UK South holds 43 resources, 82% of the inventory. This describes resource placement; it does not establish workload failover, data residency compliance or recovery capability.

Production contains the largest resource population: 45 resources, 86% of the estate. Resource counts include supporting infrastructure and are not a measure of spend or business criticality.

38 resources carry at least one tag and 14 carry none (73% with any tag). Establish the required ownership and classification tags before interpreting this as governance quality.

Decisions to make first

High-severity Defender recommendations: 1 recorded occurrences refer to 1 distinct resource IDs across 1 resolved subscriptions. Work the high-severity Defender recommendations

[High-severity Defender recommendations](#)

Management ports open to the Internet: 1 recorded occurrences refer to 1 distinct resource IDs across 1 resolved subscriptions. Close management ports to the Internet

[Management ports open to the Internet](#)

Internet-sourced network security rules: 1 recorded occurrences refer to 1 distinct resource IDs across 1 resolved subscriptions. Verify internet-sourced NSG rules

[Internet-sourced network security rules](#)

The optional technical reference contains key settings for every resource, grouped findings and collection coverage. Use the snapshot database and data exports for the underlying evidence.

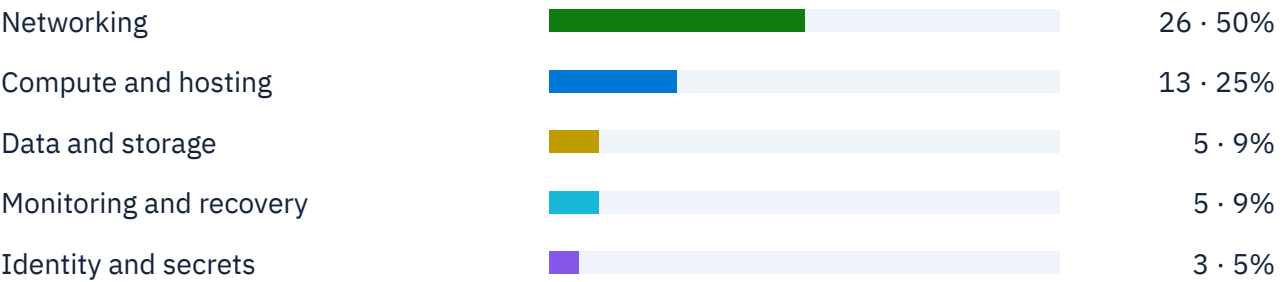
Estate composition

The comparisons below show where the estate is concentrated. Service families describe Azure resource types, including supporting components; application ownership and business purpose are not inferred from naming conventions.

Subscription comparison

Subscription	Resources	Resource groups	Affected resources
Development	7	1	7
Production	45	3	15

Service composition

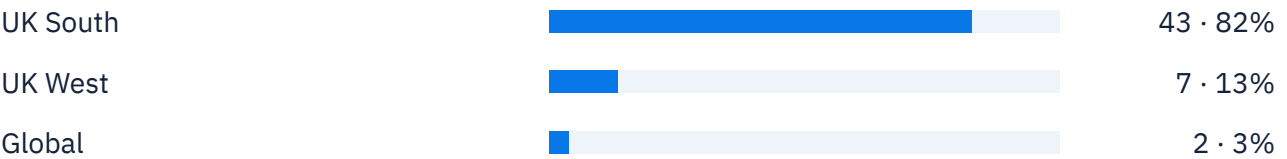


Share of recorded resources by service family. Counts include supporting infrastructure and do not measure cost or business importance.

Regional concentration



Where the estate runs. Marker area follows the number of resources in each region; global and unrecorded locations have no place on the map and appear only in the chart below.



Global and unrecorded locations remain separate. A region with few resources may still contain an important service; use the subscription profiles and dependencies to interpret the distribution.

Architecture and dependencies

The connections below are extracted from stored resource properties. Direction identifies which resource uses or attaches to another. Containment is excluded from connection counts, and reciprocal VNet peering records count as one connection. These are configuration relationships, not measured traffic flows.

34 distinct non-containment connections are recorded; 3 cross resource-group boundaries. 0 connections have at least one endpoint without a matching resource row. Missing endpoints remain explicit and do not imply a broken service.

Connection patterns

Relationship	Recorded connections
attached to	17
depends on	4
logs to	3
monitors	2
nsg attached	2
runs on	2
dns linked	1
peered with	1
private endpoint for	1
uses identity	1

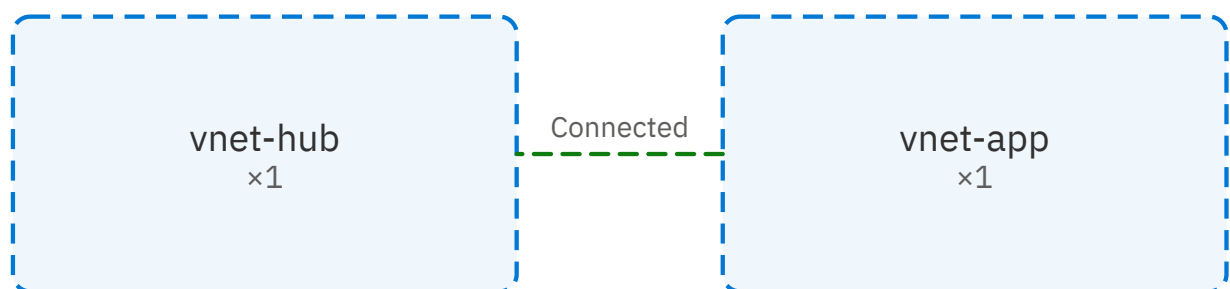
Shared dependencies

These targets have consumers in more than one resource group. Review their ownership, access and change coordination before modifying them. The count describes recorded consumers, not runtime usage or the impact of an outage.

None recorded

Network boundaries

Peering and private endpoints explain configured connectivity. A private endpoint does not by itself establish that a service has disabled public access, or that DNS and client routing use the private path.



 Virtual network

 xN aggregated resources of one type

Microsoft Azure

Virtual-network peerings recorded in this snapshot. Reciprocal records are shown once; lines describe configuration, not traffic or reachability.

Source	Target
vnet-app	vnet-hub

Private endpoints

Source	Target
pe-sql	sql-prod

Subscription profiles

Each subscription is described through its resource population, regional distribution and recorded findings. Resource-group studies are selected for finding concentration, cross-group connections and population; the technical reference contains the remaining groups and every resource. Figures illustrate up to two connection types per study, prioritising cross-group links and then connection count. Every link of a selected type is retained, with larger figures split across pages.

Development

7 resources in 1 populated resource groups; 0 resources carry at least one tag. 12 finding occurrences affect 7 distinct resources. 0 connections cross this subscription's resource-group boundaries.

rg-dev is the largest populated group with 7 resources (100% of this subscription). Its size includes attached and child resources; the study below describes the service configuration behind that count.

Service composition

Compute and hosting		4 · 57%
Identity and secrets		2 · 28%
Data and storage		1 · 14%

Share of recorded resources by service family. Counts include supporting infrastructure and do not measure cost or business importance.

Region	Resources
UK West	7

Largest resource groups

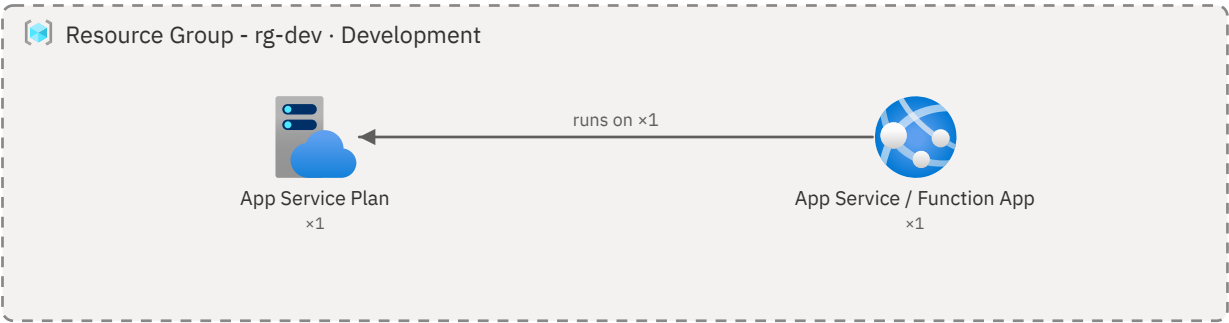
Resource group	Resources	Occurrences	Recorded connections
rg-dev	7	12	0

rg-dev

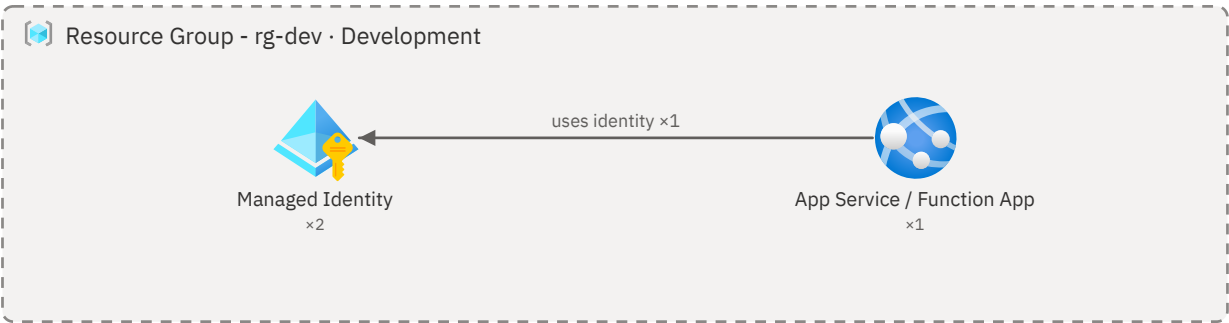
Selected for the highest remaining concentration of recorded finding severity, with high-priority occurrences considered first. This group contains 7 resources across 6 Azure types, with 12 finding occurrences and 0 connections across its boundary. 0 resources have at least one tag.

Managed Identity is the largest recorded service type, with 2 of the group's 7 resources. The composition below includes supporting and child resources; review the settings and connections to distinguish the services from their attachments.

Type	Resources
Managed Identity	2
App Service / Function App	1
App Service Certificate	1
App Service Plan	1
Disk Snapshot	1
Redis Cache	1



runs on: App Service Plan (1 recorded links) Recorded service connections for rg-dev. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



uses identity: Managed Identity (1 recorded links) Recorded service connections for rg-dev. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.

Configuration comparisons

Values are taken from the stored properties or collected inventory rows. An unrecorded value is unknown, not a default or evidence of a disabled feature.

Identity and logging configuration

Resource	Identity type	Dependency
web-dev	UserAssigned	runs on: asp-dev uses identity: id-web-dev

Dependencies across the group boundary

No cross-group dependencies are recorded. Resource Graph does not capture every application-level connection, so this does not establish isolation.

Observations to follow up

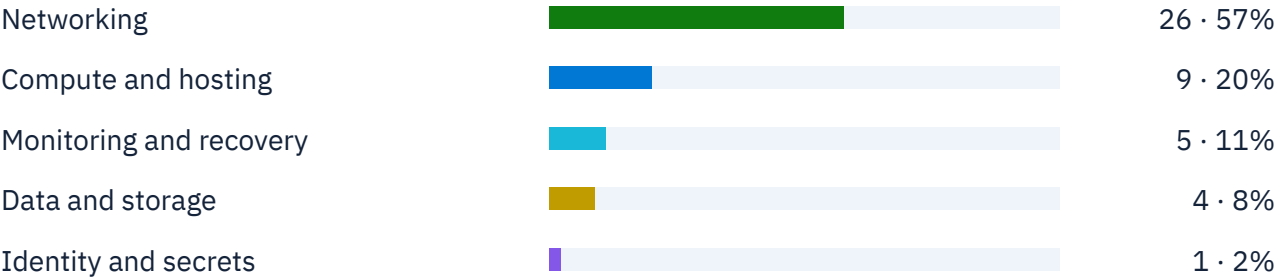
Check	Affected resources
Certificates expiring	1
Redis insecure transport	1
Web applications allowing HTTP	1
Missing required resource tags	7
Snapshots of deleted disks	1
Unreferenced managed identities	1

Production

45 resources in 3 populated resource groups; 38 resources carry at least one tag. 21 finding occurrences affect 15 distinct resources. 3 connections cross this subscription's resource-group boundaries.

rg-app is the largest populated group with 26 resources (57% of this subscription). Its size includes attached and child resources; the study below describes the service configuration behind that count.

Service composition



Share of recorded resources by service family. Counts include supporting infrastructure and do not measure cost or business importance.

Region	Resources
UK South	43
Global	2

Largest resource groups

Resource group	Resources	Occurrences	Recorded connections
rg-app	26	15	3
rg-network	18	5	2
MC_rg-app_aks-prod_uksouth	1	1	1

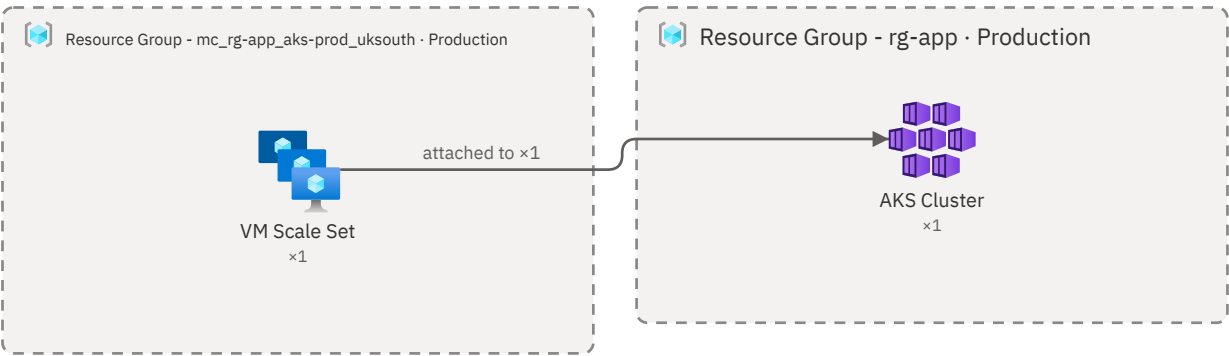
MC_rg-app_aks-prod_uksouth

Selected for the largest remaining resource population. This provides configuration context for a substantial part of the subscription.

This group contains 1 resources across 1 Azure types, with 1 finding occurrences and 1 connections across its boundary. 1 resources have at least one tag.

VM Scale Set is the largest recorded service type, with 1 of the group's 1 resources. The composition below includes supporting and child resources; review the settings and connections to distinguish the services from their attachments.

Type	Resources
VM Scale Set	1



attached to: AKS Cluster (1 recorded links) Recorded service connections for MC_rg-app_aks-prod_uksouth. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.

Configuration comparisons

Values are taken from the stored properties or collected inventory rows. An unrecorded value is unknown, not a default or evidence of a disabled feature.

No supported comparison fields are recorded for this group. The technical reference retains its resource details.

Dependencies across the group boundary

0 directed connections enter this group and 1 leave it; 0 peerings cross its boundary. Changes to the named shared targets need coordination with their consumers. These are recorded configuration links, so application dependencies may extend beyond them.

Source	Relationship	Target
aks-nodepool1-12345678-vmss	attached to	aks-prod

Observations to follow up

Check	Affected resources
Missing required resource tags	1

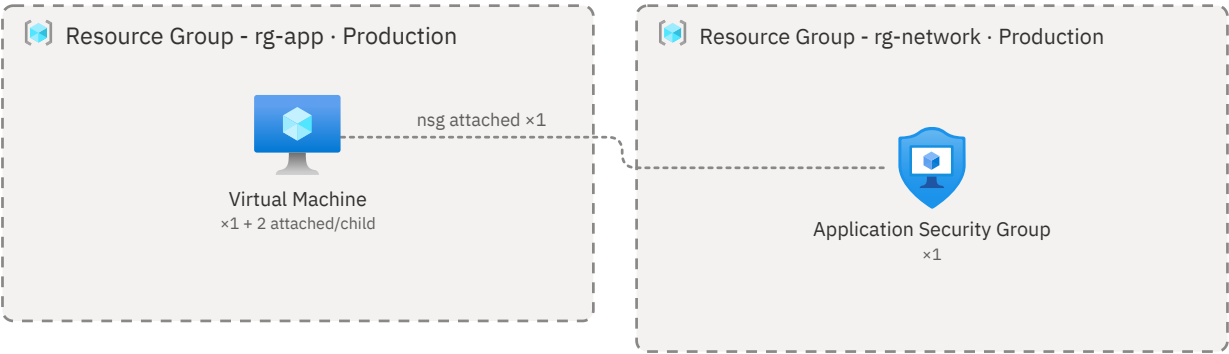
rg-app

Selected for the highest remaining concentration of recorded finding severity, with high-priority occurrences considered first. This group contains 26 resources across 24 Azure types, with 15 finding occurrences and 3 connections across its boundary. 19 resources have at least one tag.

Network Interface is the largest recorded service type, with 2 of the group’s 26 resources. The composition below includes supporting and child resources; review the settings and connections to distinguish the services from their attachments.

Type	Resources
Network Interface	2
Public IP Address	2
AKS Cluster	1
AVD Host Pool	1
Application Gateway	1
Application Gateway WAF Policy	1

Showing 6 of 24 rows; the snapshot database and data exports retain the underlying evidence.



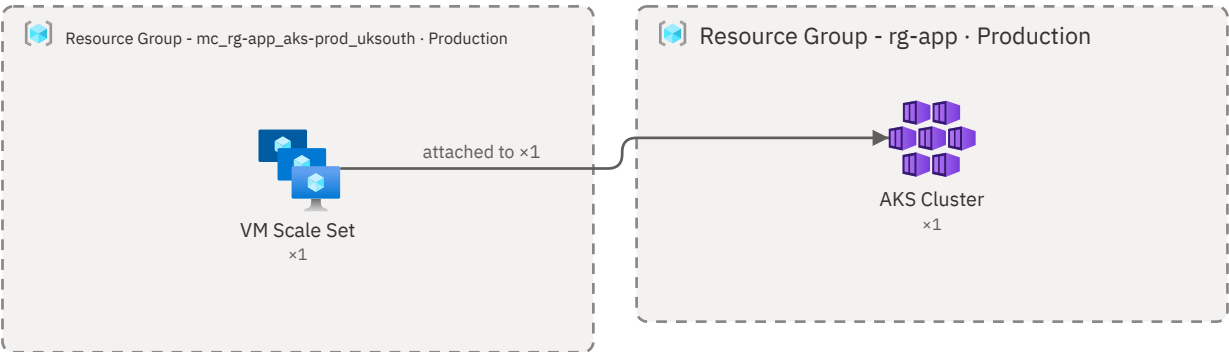
nsg attached: Application Security Group (1 recorded links) Recorded service connections for rg-app. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



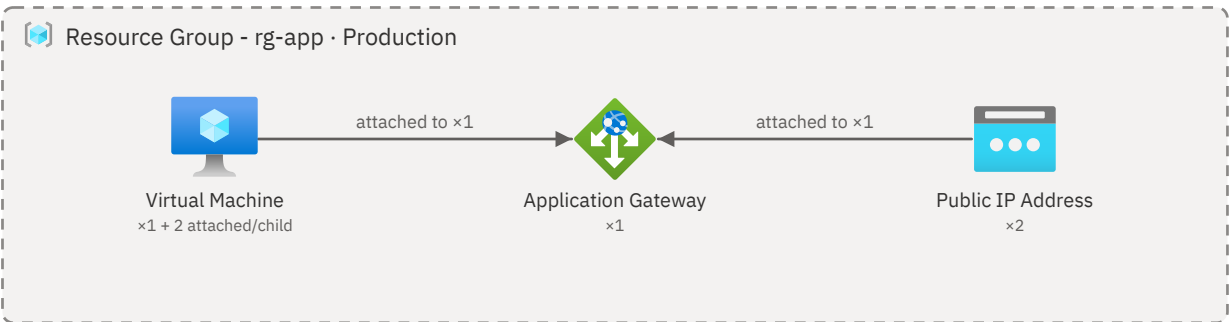
attached to: Availability Set (1 recorded links) Recorded service connections for rg-app. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: Virtual Machine (1 recorded links) Recorded service connections for rg-app. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: AKS Cluster (1 recorded links) Recorded service connections for rg-app. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: Application Gateway (2 recorded links) Recorded service connections for rg-app. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.

Configuration comparisons

Values are taken from the stored properties or collected inventory rows. An unrecorded value is unknown, not a default or evidence of a disabled feature.

Compute configuration

Resource	Size or plan	Operating system	Zones
vm-app-01	Standard_B2s	Not recorded	Not recorded

Storage configuration

Resource	SKU or replication	Public network access	Anonymous access permitted
stprodapp01	Not recorded	Not recorded	true

Database network configuration

Resource	Public network access	Default network action
psql-prod	Disabled	Not recorded
sql-prod	Disabled	Not recorded

Identity and logging configuration

Resource	Identity type	Dependency
ca-api	Not recorded	runs on: cae-prod
des-prod	SystemAssigned	None recorded
aks-prod	SystemAssigned	logs to: law-prod
appi-prod	Not recorded	logs to: law-prod
dcr-prod	Not recorded	logs to: law-prod

Dependencies across the group boundary

1 directed connections enter this group and 1 leave it; 1 peerings cross its boundary. Changes to the named shared targets need coordination with their consumers. These are recorded configuration links, so application dependencies may extend beyond them.

Source	Relationship	Target
vnet-app	peered with	vnet-hub
vm-app-01-nic	nsg attached	asg-web
aks-nodepool1-12345678-vmss	attached to	aks-prod

Observations to follow up

Check	Affected resources
High-severity Defender recommendations	1
Storage anonymous-access permission	1
Storage accounts open to all networks	1
AKS local accounts enabled	1
Retired Basic SKU networking	1
SQL TLS floor below 1.2	1
Missing required resource tags	7
SQL authentication still accepted	1
Shared key authorisation permitted	1

rg-network

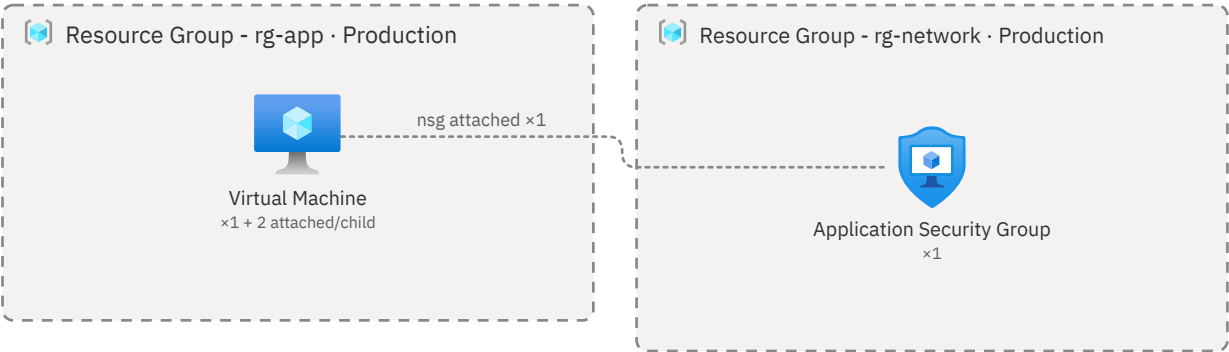
Selected for the greatest remaining number of recorded cross-group connections. This highlights dependency coordination rather than assuming business criticality.

This group contains 18 resources across 13 Azure types, with 5 finding occurrences and 2 connections across its boundary. 18 resources have at least one tag.

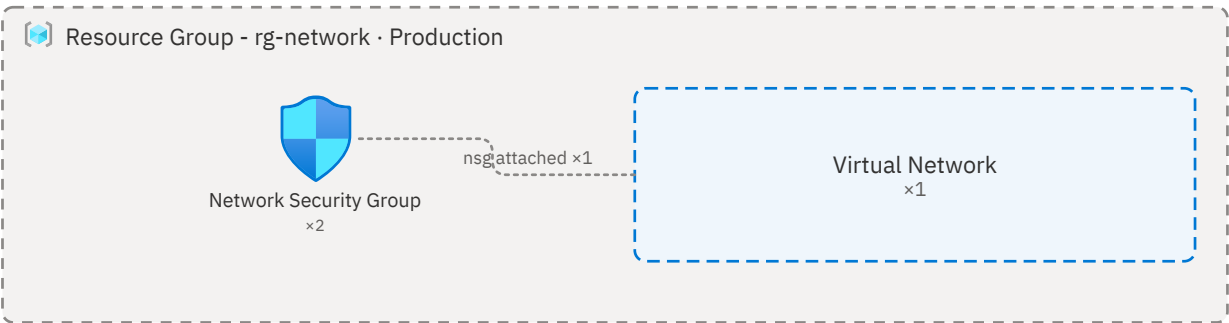
Public IP Address is the largest recorded service type, with 3 of the group’s 18 resources. The composition below includes supporting and child resources; review the settings and connections to distinguish the services from their attachments.

Type	Resources
Public IP Address	3
Network Security Group	2
Private DNS Zone	2
Route Table	2
Application Security Group	1
Azure Firewall	1

Showing 6 of 13 rows; the snapshot database and data exports retain the underlying evidence.



nsg attached: Application Security Group (1 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



Virtual network xN aggregated resources of one type [Microsoft Azure](#)

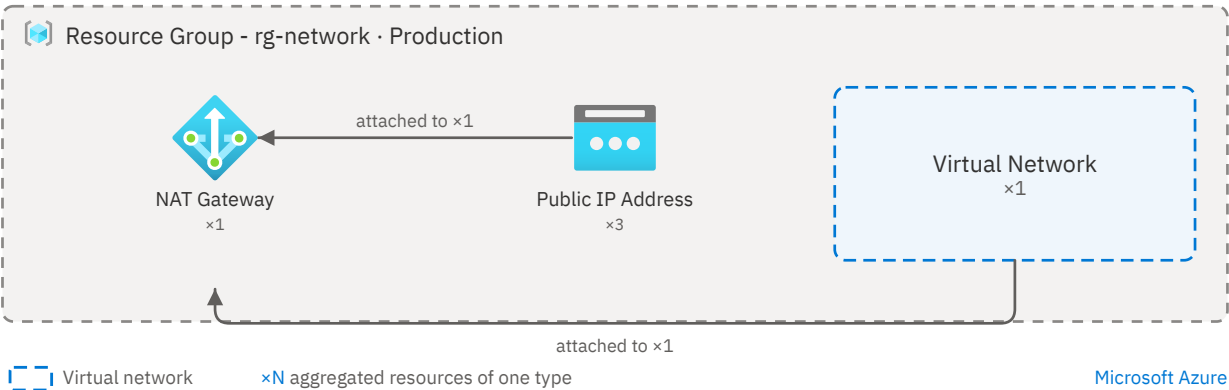
nsg attached: Network Security Group (1 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



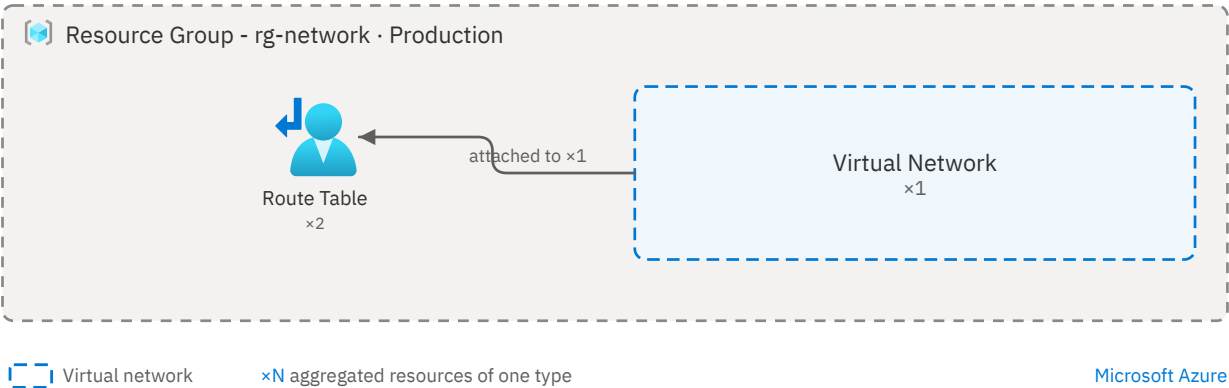
attached to: Azure Firewall (1 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



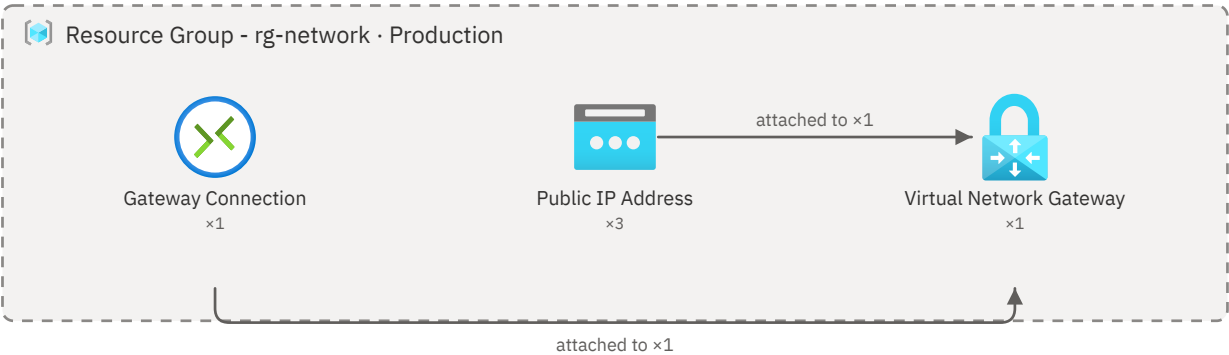
attached to: Local Network Gateway (1 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: NAT Gateway (2 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: Route Table (1 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.



attached to: Virtual Network Gateway (2 recorded links) Recorded service connections for rg-network. Attached NICs and disks fold into their host where possible; repeated types are aggregated with counts. External endpoints show dependencies outside this group. Lines count observed links, not every member of an aggregate.

Configuration comparisons

Values are taken from the stored properties or collected inventory rows. An unrecorded value is unknown, not a default or evidence of a disabled feature.

No supported comparison fields are recorded for this group. The technical reference retains its resource details.

Dependencies across the group boundary

1 directed connections enter this group and 0 leave it; 1 peerings cross its boundary. Changes to the named shared targets need coordination with their consumers. These are recorded configuration links, so application dependencies may extend beyond them.

Source	Relationship	Target
vnet-app	peered with	vnet-hub
vm-app-01-nic	nsg attached	asg-web

Observations to follow up

Check	Affected resources
Management ports open to the Internet	1
Internet-sourced network security rules	1
Private DNS zones without links	1
Route tables without subnets	1
Network security groups without an association	1

Audit findings and review priorities

Each recurring check is explained once. Severity labels are retained from collection and describe the audit rule; the assessment distinguishes observed settings, potential exposure and policy choices. Confirm the affected service configuration before making changes.

High-severity Defender recommendations

HIGH Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
High	1

What was observed

The audit recorded Defender for Cloud recommendations that were in an unhealthy state at high severity for the affected resource.

Why it matters

Defender's own assessment already ranks these as the most consequential gaps. They are recommendations, not confirmed exploits, and depend on the plans enabled in each subscription.

What to verify

Open each recommendation in Defender for Cloud, confirm the affected resource and its remediation steps, and record exemptions where the finding is accepted.

Policy considerations

High-severity Defender recommendations should have an owner and a target date; exemptions should be recorded in Defender, not in a spreadsheet.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- vm-app-01: high severity Defender assessment is unhealthy

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/defender-for-cloud/review-security-recommendations>

Management ports open to the Internet

HIGH Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
High	1

What was observed

The audit recorded inbound NSG rules that allow RDP (3389) or SSH (22) from any Internet source, including rules written with address and port arrays.

Why it matters

Exposed management ports are the most common brute-force and credential-stuffing target on Azure. A permissive rule does not prove a listener exists behind it, but it removes the network boundary that would otherwise contain one.

What to verify

Confirm which NICs and subnets the NSG protects and whether a VM listens on the port. Replace Internet-wide rules with Bastion, just-in-time access or a source-restricted rule, then re-test the intended administrative path.

Policy considerations

Administrative access should terminate on a bastion or VPN, never on a public listener. Record any approved exception with its source restriction and review date.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-network (Production), the largest resolved group concentration for this check.

Evidence examples

- nsg-app: rule allow-ssh opens management port 22 to the Internet

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/virtual-network/network-security-groups-overview>

Internet-sourced network security rules

HIGH Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
High	1

What was observed

The audit recorded network security rules matching the audit's internet-access criteria.

Why it matters

A matching rule may permit internet-sourced traffic, but priority, association, destination and other controls determine the effective path. The stored rule alone does not establish end-to-end reachability.

What to verify

Review rule priority, direction, ports, source ranges and effective rules on associated NICs and subnets.

Policy considerations

Document permitted public entry points and restrict administrative access to approved paths; verify dependencies before changing rules.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-network (Production), the largest resolved group concentration for this check.

Evidence examples

- nsg-app: rule allow-ssh allows Internet -> port 22

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source: the original stored finding and its collection-time query evidence.

Storage anonymous-access permission

HIGH Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
High	1

What was observed

The audit recorded storage accounts with the account-level permission for anonymous blob access enabled.

Why it matters

This permits containers to be configured for anonymous access. It does not establish that any container or blob is currently public; container access level and network controls also matter.

What to verify

Inspect container access levels and known anonymous consumers. Disable the account permission where anonymous delivery is not intended, after validating dependent services.

Policy considerations

Public content delivery may be intentional. Document its scope and separate it from confidential data rather than treating every permitted account as a confirmed disclosure.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- stprodapp01 allows public blob access

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/storage/blobs/anonymous-read-access-configure>

Storage accounts open to all networks

HIGH Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
High	1

What was observed

The audit recorded storage accounts whose public network access is enabled and whose firewall default action allows every source.

Why it matters

The account accepts requests from any network; authentication, shared-key settings and container access levels still decide what those requests can do. This is the widest network posture a storage account can have.

What to verify

Review the clients that reach each account, then move to selected networks, private endpoints or trusted-service exceptions. Test the dependent workloads before changing the default action to deny.

Policy considerations

Storage accounts holding regulated or internal data should default to deny with explicit allowances. Public delivery accounts can stay open but should be documented as such.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- stprodapp01 allows public network access

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/storage/common/storage-network-security>

AKS local accounts enabled

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded AKS clusters whose Kubernetes local accounts remain enabled, or where the setting was not present and therefore defaults to enabled.

Why it matters

Local accounts provide a static cluster-admin credential outside Entra. With Entra integration on, they are a bypass; with it off, they are the only path and the cluster has no identity-backed access at all.

What to verify

Enable Entra integration if it is not already on, move administrators to Entra groups with Azure RBAC for Kubernetes, then disable local accounts and rotate any previously issued kubeconfig.

Policy considerations

Production clusters should run Entra-only access; break-glass access should be documented rather than left permanently enabled.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- aks-prod allows local Kubernetes accounts

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/aks/manage-local-accounts-managed-azure-ad>

Retired Basic SKU networking

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded public IP addresses and load balancers on the Basic SKU, which Azure retired on 30 September 2025.

Why it matters

Retired SKUs receive no support and may stop working at Microsoft's discretion. Basic SKUs also lack availability zones and the Standard security-by-default posture.

What to verify

Plan the migration to Standard SKU following Microsoft's upgrade guidance, which changes the default inbound posture and may need NSG rules added.

Policy considerations

No new Basic SKU resources should be created; existing ones need a migration owner and date.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- vm-app-01-pip uses the retiring Basic SKU

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/load-balancer/skus>

Certificates expiring

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded App Service certificates that had expired, or were due to expire within 30 days of the collection time.

Why it matters

An expired certificate breaks TLS for every hostname bound to it. Auto-renewal covers managed certificates, but imported and Key Vault certificates depend on someone renewing them.

What to verify

Confirm whether the certificate is still bound, whether renewal is automated, and renew or retire it before the expiry date. Re-collect afterwards to clear the observation.

Policy considerations

Track certificate ownership and renewal dates outside the platform; treat an expiry inside the review window as a scheduled task, not an incident.

Where it is concentrated

Subscription	Affected resources
Development	1

1 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- web-dev-cert expires within 30 days

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/app-service/configure-ssl-certificate>

Defender plans on the Free tier

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 1.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded Defender for Cloud plans set to the Free tier in a subscription, meaning that workload type has no threat protection and its assessments are limited to the foundational set.

Why it matters

Every other Defender observation in this report is bounded by which plans are on. A plan on the Free tier is a blind spot, not a finding of safety.

What to verify

Decide per subscription which plans the workloads warrant, enable them, and re-collect to see the assessments and alerts they produce.

Policy considerations

Plan coverage is a budget decision; record which plans are deliberately off and why so the gap is understood rather than rediscovered.

Where it is concentrated

Evidence examples

- Defender for Servers is off in sub-prod

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/defender-for-cloud/enable-all-plans>

Key-based local authentication permitted

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 1.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded Cognitive Services and AI Search accounts that still accept API keys, including accounts where the setting was not present and therefore defaults to permitted.

Why it matters

API keys are long-lived, unscoped secrets that bypass conditional access. Disabling local authentication forces Entra identities and role assignments.

What to verify

Find the callers using keys, migrate them to managed identities with the appropriate data-plane roles, then disable local authentication and confirm the callers still succeed.

Policy considerations

AI and search endpoints handling internal data should be Entra-only; public demo endpoints may keep keys with a documented scope.

Where it is concentrated

Evidence examples

- oai-prod accepts local (key) authentication

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/ai-services/authentication>

Redis insecure transport

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded Redis caches with the non-SSL port enabled or a minimum TLS version below 1.2.

Why it matters

The non-SSL port carries credentials and cached data in clear text; an old TLS floor allows weak negotiation. Neither proves a client uses them, but both keep the option open.

What to verify

Check client connection strings for port 6379 and old TLS settings, switch them to 6380 with TLS 1.2, then disable the non-SSL port.

Policy considerations

Caches should expose only the TLS port with a 1.2 floor; exceptions need an owner and a date.

Where it is concentrated

Subscription	Affected resources
Development	1

1 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- redis-dev accepts non-SSL connections

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/azure-cache-for-redis/cache-remove-tls-10-11>

SQL TLS floor below 1.2

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded SQL servers or managed instances whose minimal TLS version explicitly permits 1.0, 1.1 or none.

Why it matters

Older TLS versions have known weaknesses and are disabled by most current clients. The setting only matters for clients that would negotiate below 1.2, but it removes the guarantee that no connection does.

What to verify

Confirm no legacy client still negotiates old TLS, then raise the floor to 1.2 and re-test connectivity from every application.

Policy considerations

A 1.2 floor is the platform baseline; anything lower should be a documented exception with a retirement plan.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- sql-prod accepts TLS 1.0

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/azure-sql/database/connectivity-settings>

Web applications allowing HTTP

MEDIUM Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Medium	1

What was observed

The audit recorded web applications that do not enforce HTTPS-only access.

Why it matters

Applications may accept requests over HTTP unless another layer redirects or blocks them. Review the complete ingress path, including proxies and custom domains.

What to verify

Test HTTP and HTTPS behaviour, redirects and dependent clients. Enable HTTPS-only where required after confirming certificates and client compatibility.

Policy considerations

Define transport requirements for public and private application endpoints.

Where it is concentrated

Subscription	Affected resources
Development	1

1 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- web-dev does not enforce HTTPS-only traffic

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/app-service/configure-ssl-bindings>

Missing required resource tags

LOW Recorded occurrences: 15. Distinct resource IDs: 15. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Low	15

What was observed

The audit recorded resources missing at least one tag required by the collection-time audit.

Why it matters

Missing classification keys make ownership and cost allocation harder to establish. A present key may still hold an empty or unsuitable value, so key presence alone does not establish useful classification.

What to verify

Use the stored missing-key evidence to identify owners and correct omissions; do not substitute today's tag policy.

Policy considerations

Use the collection-time missing keys as evidence and confirm the current required keys, accepted values and exception process with their owners.

Where it is concentrated

Subscription	Affected resources
Development	7
Production	8

7 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- redis-dev is missing tags: env
- web-dev-snap is missing tags: env
- id-unused is missing tags: env

Showing 3 of 15 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source: the original stored finding and its collection-time query evidence.

SQL authentication still accepted

LOW Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Low	1

What was observed

The audit recorded SQL servers that still accept SQL logins alongside Microsoft Entra authentication, or that have no Entra-only setting recorded.

Why it matters

SQL logins are passwords stored outside the identity platform, without conditional access or MFA. Entra-only authentication removes that surface entirely.

What to verify

Identify workloads still using SQL logins, move them to managed identities or Entra service principals, then enable Entra-only authentication.

Policy considerations

Entra-only authentication should be the default for new servers; legacy dependencies should be listed with owners.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- sql-prod does not require Entra-only authentication

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source

<https://learn.microsoft.com/en-us/azure/azure-sql/database/authentication-azure-ad-only-authentication>

Shared key authorisation permitted

LOW Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Low	1

What was observed

The audit recorded storage accounts that still accept account-key (shared key) authorisation, including accounts where the setting was not present and therefore defaults to permitted.

Why it matters

Account keys grant full data-plane access without identity, cannot be scoped and are a common leak vector in pipelines and configuration files. Disabling them forces Entra-based authorisation.

What to verify

Inventory the consumers of each account's keys and SAS tokens, migrate them to managed identities or user delegation SAS, then disable shared key access and confirm the workloads still run.

Policy considerations

New accounts should disable shared key access from the start. Existing accounts should carry a migration date or a documented exception.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-app (Production), the largest resolved group concentration for this check.

Evidence examples

- stprodapp01 allows shared key access

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/storage/common/shared-key-authorization-prevent>

Snapshots of deleted disks

INFO Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Info	1

What was observed

The audit recorded disk snapshots whose source disk no longer exists in the estate.

Why it matters

Such snapshots cost storage indefinitely and are rarely restorable to their original purpose. They may still be an intentional archive.

What to verify

Confirm whether the snapshot backs an image or a retention requirement; delete or move it to a documented archive otherwise.

Policy considerations

Snapshot retention should be a policy with a lifetime, not the default of forever.

Where it is concentrated

Subscription	Affected resources
Development	1

1 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- web-dev-snap's source disk no longer exists

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/virtual-machines/snapshot-copy-managed-disk>

Private DNS zones without links

INFO Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Info	1

What was observed

The audit recorded private DNS zones linked to no virtual network.

Why it matters

A zone with no links resolves nothing; it may be a leftover, a zone awaiting a link, or one used only for private-endpoint registration that has since been removed.

What to verify

Check whether a private endpoint or workload still expects the zone, link it or remove it.

Policy considerations

Zone ownership should be recorded alongside the private endpoints that use it.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-network (Production), the largest resolved group concentration for this check.

Evidence examples

- privatelink.blob.core.windows.net is linked to no virtual network

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source

<https://learn.microsoft.com/en-us/azure/dns/private-dns-virtual-network-links>

Route tables without subnets

INFO Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Info	1

What was observed

The audit recorded route tables associated with no subnet.

Why it matters

A detached route table affects nothing; it may be a leftover from a redesign or a table prepared for a subnet not yet created.

What to verify

Confirm intent with the network owner and remove tables that no design references.

Policy considerations

Network artefacts should be removed with the design that created them.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-network (Production), the largest resolved group concentration for this check.

Evidence examples

- rt-unused is associated with no subnet

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/virtual-network/manage-route-table>

Network security groups without an association

INFO Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Info	1

What was observed

The audit recorded network security groups without a NIC or subnet association.

Why it matters

An unassociated NSG is not enforcing rules on those resources. It may be intentionally staged or left behind; the record does not establish that another resource lacks protection.

What to verify

Confirm whether the NSG is referenced by deployment templates or planned changes. Identify an owner before attaching or removing it.

Policy considerations

Review lifecycle and deployment ownership; do not infer cost savings from this inventory alone.

Where it is concentrated

Subscription	Affected resources
Production	1

1 affected resources are in rg-network (Production), the largest resolved group concentration for this check.

Evidence examples

- nsg-unused is not associated with any subnet or NIC

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/azure/virtual-network/network-security-groups-overview>

Unreferenced managed identities

INFO Recorded occurrences: 1. Distinct resource IDs: 1. Estate-level occurrences: 0; occurrences without a matching resource row: 0.

Recorded severity mix	Occurrences
Info	1

What was observed

The audit recorded user-assigned managed identities that no collected resource references. Federated credentials and AKS kubelet identities are not visible to this check.

Why it matters

An identity nobody uses may still hold role assignments, which is standing privilege with no consumer. It may also be used by a resource type outside the inventory.

What to verify

Check role assignments and federated credentials on each identity, confirm no workload uses it, then remove it and its assignments.

Policy considerations

Identities should be created with their consumer and removed with it.

Where it is concentrated

Subscription	Affected resources
Development	1

1 affected resources are in rg-dev (Development), the largest resolved group concentration for this check.

Evidence examples

- id-unused is assigned to no resource

Showing 1 of 1 occurrences. The technical reference groups all occurrences by resource; original titles and stored evidence remain in the snapshot database and data exports.

Source <https://learn.microsoft.com/en-us/entra/identity/managed-identities-azure-resources/overview>

Governance and operational evidence

Tag presence supports resource classification, but any-tag coverage is not an ownership, cost-allocation or compliance verdict. Required-tag findings describe the audit performed at collection time; today's configuration is not reapplied to an old snapshot.

38 of 52 resources carry any tag (73%). The application uses 60% as a tag-presence threshold only; passing it does not demonstrate that required tags or useful values exist.

Development has the lowest any-tag coverage at 0%. Start by confirming the required keys and the teams responsible for the untagged resources, then review coverage at resource-group level.

Coverage by subscription

Subscription	Tag coverage
Development	0%
Production	84%

Coverage by tag key

Share of the 38 tagged resources carrying each key.

Tag key	Resources	Share
env	37	97%
aks-managed-clusterName	1	2%
aks-managed-poolName	1	2%

15 resources have stored required-tag findings. The missing keys below are historical audit evidence; they do not establish the complete policy that was configured.

Resource group	Non-compliant	Missed tags
Production / rg-app	7	env
Development / rg-dev	7	env
Production / MC_rg-app_aks-prod_uksouth	1	env

Monitoring and recovery evidence

1 logging destinations receive recorded connections from 3 distinct resources. 2 monitoring connections and 1 recovery or backup vaults are recorded.

Logging connections, monitoring resources and recovery vaults show installed or configured infrastructure. This snapshot does not contain backup job outcomes, restore tests, alert delivery tests, application health or utilisation. Assign those checks separately before drawing an operational assurance conclusion.

Logging targets	Distinct consumers
law-prod	3

Vault configuration	Location
rsv-prod	UK South

Operational checks to complete

Confirm which services require central logging, verify that telemetry arrives and alerts reach an accountable team, and record the last successful restore test for each recovery-critical workload. Review retention and replication against the organisation's own recovery objectives.

Operational and compliance evidence

Microsoft service evidence recorded during collection. Results cover only the accessible scope; an empty dataset is not proof of zero cost, compliance, protection or service enablement. Detailed records remain in the snapshot database and data exports.

Azure Advisor cost opportunities

Currency	Period	Items	Savings	Annual savings
GBP	Month	1	125.50	Unknown

Collection: Recorded evidence

Estimates from Azure Advisor, grouped by currency and reported period. Annual estimates are separate. Missing amounts or periods remain unknown. Recommendations can overlap; totals are not guaranteed savings or billed costs.

Azure Policy evaluation states

Sub. ID	Assignment	Initiative ID	State	Results
sub-prod	/subscriptions/sub-prod/providers/microsoft.authorization/policyassignments/baseline	/providers/microsoft.authorization/policysetdefinitions/security-baseline	Exempt	1

Collection: Recorded evidence

Counts are policy evaluations, not distinct resources. Recorded states stay separate; missing evaluations never imply compliance. Full assignment and initiative IDs identify each group. A missing initiative may indicate a standalone policy or missing evidence.

Azure Policy exemptions

Category	Expiry at collection	Exemptions
Waiver	Due after 90 days	1

Collection: Recorded evidence

Expiry is assessed at collection time. Review expired exemptions and those due within 90 days with their owners. No expiry specified does not mean permanent approval; assignment scope and justification remain in the stored evidence.

Defender regulatory standards

Standard	State	Standard records
Azure-Security-Benchmark	Failed	1

Collection: Recorded evidence

Standard states across accessible subscriptions. Detailed control counts remain in the stored evidence. Missing or unsupported coverage does not establish compliance or certification.

Defender regulatory controls

Standard	State	Control records
Azure-Security-Benchmark	Unsupported	1

Collection: Recorded evidence

Control states across accessible subscriptions, grouped by standard. Skipped, unsupported and unknown states remain distinct from passes. Full control IDs remain in the stored evidence.

Defender regulatory assessments

Standard	State	Checks	Passed	Failed	Skipped
Azure-Security-Benchmark	Failed	1	12	2	Unknown

Collection: Recorded evidence

Passed, failed and skipped counts are resource occurrences across assessments; resources can be counted more than once. Missing counts remain unknown. These assessments do not establish certification or compliance of unassessed resources.

Policy assignment inventory

Enforcement	Assignments
Default	1
DoNotEnforce	1

Collection: Recorded evidence

Assignment inventory includes the requested inherited scope. Enforcement mode and excluded scopes affect interpretation; an assignment does not establish a passing evaluation.

Patch assessments

OS	State	Checks	Security updates	Critical updates
Linux	Succeeded	1	2	Unknown

Collection: Recorded evidence

Counts describe pending updates at the recorded assessment time and may overlap across classifications. Missing counts remain unknown. ARG assessment history is limited to seven days.

Backup protection evidence

Collection: Evidence incomplete or collection status unverified

Protection state, backup status and recovery-point time are separate evidence. Missing fields remain unknown; the age threshold is a configurable review aid, not a workload RPO.

Backup and restore outcomes

Operation	State	Jobs
Backup	CompletedWithWarnings	1
Restore	Failed	1

Collection: Recorded evidence

Backup and restore jobs retain the provider status, including warnings. ARG exposes up to 14 days of jobs; an empty result does not prove successful backups.

Recent ARM changes

Type	Actor type	Events
Update	Application	1

Collection: Recorded evidence

Recorded ARM changes can include actor, client and changed properties. ARG retains 14 days; this is not a complete activity or data-plane audit log.

Policy evaluation coverage

Assignment	Name	State	Results
/subscriptions/sub-prod/providers/microsoft.authorization/policyassignments/baseline	Production baseline	Evaluation observed	1
/subscriptions/sub-prod/providers/microsoft.authorization/policyassignments/not-evaluated	Pending initiative	No evaluation observed	0

Collection: Recorded evidence

Assignments are matched to stored policy states using full normalised ARM IDs. No observed evaluation is kept separate from compliance and from unavailable state collection.

Machine assessment coverage

Type	State	Resources
microsoft.compute/virtualmachines	Evaluation observed	1

Collection: Recorded evidence

Counts use the collected VM and Arc inventory, deduplicated by full ARM ID. No observed assessment may indicate missing service configuration, access or expired evidence; it is not a passed patch check.

Evidence age at collection

Query	Review age (hours)	State	Items
patch_assessments	72	Missing or invalid timestamp	1

Collection: Recorded evidence

Ages are evaluated at the snapshot timestamp using the threshold recorded with each query. Future, invalid and missing dates stay separate. Review thresholds are configurable and are not Microsoft retention guarantees.

Website screenshots

Saved website views from this machine. Capture time is separate from the Azure audit time.

Resource	Website	Status	Captured
web-dev	https://web-dev.azurewebsites.net/	Not captured	
agw-prod	listener	No hostname configured	

Prioritised review actions

Use this register to organise review work. The order preserves recorded severity, then the number of affected resources; it is not a business-impact score or an automatic remediation schedule. Establish owners and change dependencies before implementation.

Recommended review	Severity	Affected resources
Work the high-severity Defender recommendations	High	1
Close management ports to the Internet	High	1
Verify internet-sourced NSG rules	High	1
Verify storage anonymous-access requirements	High	1
Restrict storage network access	High	1
Disable AKS local accounts	Medium	1
Migrate Basic SKU networking to Standard	Medium	1
Renew expiring certificates	Medium	1
Review Defender plan coverage	Medium	1
Retire API-key authentication	Medium	1
Disable the Redis non-SSL port	Medium	1
Raise the SQL TLS floor	Medium	1
Verify application HTTPS enforcement	Medium	1
Resolve historical required-tag omissions	Low	15
Move SQL access to Entra-only authentication	Low	1
Retire shared key authorisation	Low	1
Review orphaned snapshots	Info	1
Review unlinked private DNS zones	Info	1
Review detached route tables	Info	1
Confirm the purpose of unassociated NSGs	Info	1
Review unreferenced managed identities	Info	1

Operational checks to complete

Collection coverage

The report describes the selected stored snapshot. Successful zero-row queries, query failures and missing run evidence are different states. A completed collection does not establish that every Azure service or control was assessed.

Snapshot status: complete. 11 queries have successful run records, including 0 with zero rows; 0 failed and 5 have no recorded outcome.

Snapshot

a7de1a01-f9b1-4ad0-a4c6-06820e65e8f3

Collected

2026-09-22T21:38:37.444857+00:00

Tenant

fixture-tenant

Collected query	Outcome	Rows
Defender compliance assessments	Completed	1
Defender compliance controls	Completed	1
Defender compliance standards	Completed	1
Advisor cost recommendations	Completed	1
Policy assignments	Completed	2
Policy exemptions	Completed	1
Policy states	Completed	1
All resources	Completed	52
Resource changes	Completed	1
Patch assessments	Completed	1
Backup jobs	Completed	2

The collector observes resource configuration through Azure Resource Graph. Effective user access, application traffic, billing, runtime health, backup success and recovery testing require additional evidence. Missing queries or resource properties must not be interpreted as a passed control.

This guidance explains the check's subject. The original recorded evidence remains authoritative, including when a locally overridden query uses the same check name.